MAT 110

Lecture 06

Increasing and Decreasing Functions

Critical Points, Relative extrema values

Akram Hosen (RMH)
Lecturer, MPS.
BRAC University

Contents

1 Increasing, Decreasing and Constant Functions

Definition: Increasing, decreasing and constant functions

Let f be defined on an interval I, and let $x_1, x_2 \in I$ with $x_1 < x_2$.

(a) The function f is called **increasing** on I if

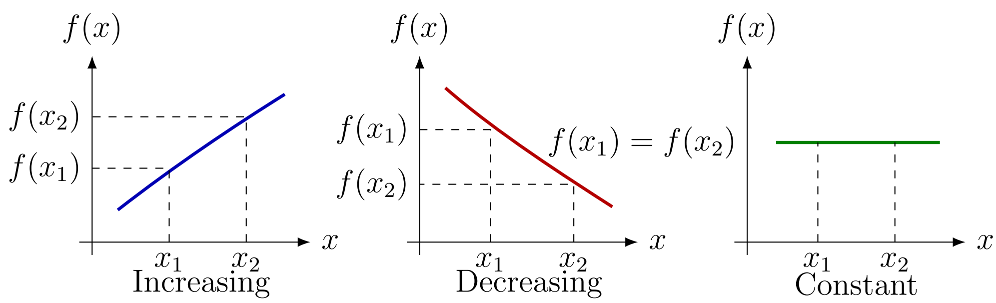
$$f(x_1) < f(x_2) \quad \text{whenever} \quad x_1 < x_2.$$

(b) The function f is called **decreasing** on I if

$$f(x_1) > f(x_2) \quad \text{whenever} \quad x_1 < x_2.$$

(c) The function f is called **constant** on I if

$$f(x_1) = f(x_2) \quad \text{whenever} \quad x_1 < x_2.$$



Graphical idea

When a graph rises as we move from left to right, the function is increasing. When it falls from left to right, the function is decreasing. If the graph stays horizontal, the function is constant.

2 Derivative Test for Increasing and Decreasing Functions

Theorem: First derivative test for monotonicity

Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b).

(a) If $f'(x) > 0$ for every $x \in (a, b)$, then f is increasing on $[a, b]$.

(b) If $f'(x) < 0$ for every $x \in (a, b)$, then f is decreasing on $[a, b]$.

(c) If $f'(x) = 0$ for every $x \in (a, b)$, then f is constant on $[a, b]$.

Increasing and Decreasing Intervals

Let f be differentiable on an interval I.

(a) If $f'(x) > 0$ for all $x \in I$, then f is increasing on I.

(b) If $f'(x) < 0$ for all $x \in I$, then f is decreasing on I.

3 Finding Intervals of Increase and Decrease

Standard procedure

To find the intervals on which a differentiable function is increasing or decreasing:

1. Find $f'(x)$.
2. Solve $f'(x) = 0$ and also note where $f'(x)$ is undefined.
3. Use these points to split the domain into intervals.
4. Test the sign of $f'(x)$ on each interval.
5. If $f'(x) > 0$, the function is increasing; if $f'(x) < 0$, the function is decreasing.

Example 1: Intervals for $y = x^3$

Find the intervals on which

$$y = x^3$$

is increasing or decreasing.

Answer.

Given

$$f(x) = x^3.$$

Differentiate:

$$f'(x) = 3x^2.$$

Now examine the sign of $f'(x)$:

$$\begin{aligned} 3x^2 > 0 &\quad \text{for} \quad x < 0, \\ 3x^2 = 0 &\quad \text{at} \quad x = 0, \\ 3x^2 > 0 &\quad \text{for} \quad x > 0. \end{aligned}$$

Thus $f'(x)$ is positive on $(-\infty, 0)$ and $(0, \infty)$, and it is zero only at $x = 0$. Therefore, $f(x) = x^3$ is increasing on

$$(-\infty, 0] \quad \text{and} \quad [0, \infty).$$

Since the function is continuous everywhere and there is no interval where $f'(x) < 0$, the function is increasing on the whole real line:

$$\boxed{(-\infty, \infty)}.$$

There is no decreasing interval.

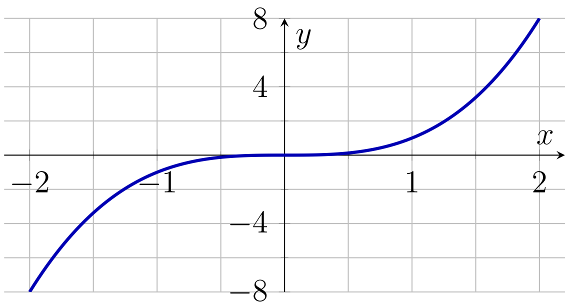


Figure: The graph of $y = x^3$ is increasing from left to right.

Example 2: Intervals for a polynomial

Find the intervals on which

$$f(x) = 3x^4 + 4x^3 - 12x^2 + 2$$

is increasing or decreasing.

Answer.

Given$f(x) = 3x^4 + 4x^3 - 12x^2 + 2$. Differentiate:

$$\begin{aligned} f'(x) &= 12x^3 + 12x^2 - 24x \\ &= 12x(x^2 + x - 2) \\ &= 12x(x + 2)(x - 1). \end{aligned}$$

The critical numbers obtained from $f'(x) = 0$ are

$$12x(x + 2)(x - 1) = 0 \quad \Longrightarrow \quad x = -2, \ 0, \ 1.$$

These numbers divide the real line into four intervals:

$$(-\infty, -2), \quad (-2, 0), \quad (0, 1), \quad (1, \infty).$$

Now check the sign of

$$f'(x) = 12x(x + 2)(x - 1).$$

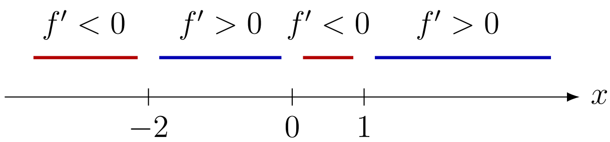
Interval	Sign of $x(x+2)(x-1)$	Sign of $f'(x)$	Conclusion
$x < -2$	$(-)(-)(-) = -$	$-$	Decreasing on $(-\infty, -2)$
$-2 < x < 0$	$(-)(+)(-) = +$	$+$	Increasing on $(-2, 0)$
$0 < x < 1$	$(+)(+)(-) = -$	$-$	Decreasing on $(0, 1)$
$x > 1$	$(+)(+)(+) = +$	$+$	Increasing on $(1, \infty)$

Therefore,

$$\boxed{f \text{ is increasing on } (-2,0) \cup (1,\infty)}$$

and

$$\boxed{f \text{ is decreasing on } (-\infty,-2) \cup (0,1)}.$$



Sign chart for $f'(x) = 12x(x+2)(x-1)$.

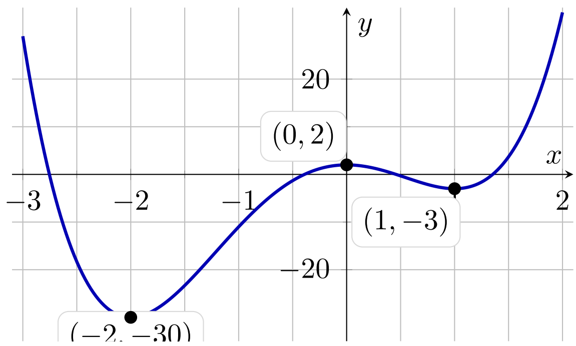


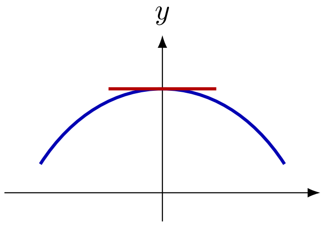
Figure: The graph of $y = 3x^4 + 4x^3 - 12x^2 + 2$.

The graph confirms decreasing, increasing, decreasing and increasing behaviour in the four intervals.

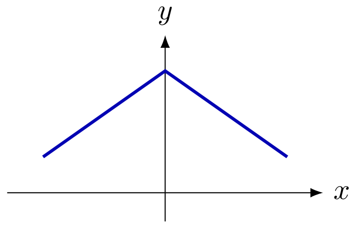
4 Critical Points

Definition: Critical point

A point x_0 in the domain of f is called a **critical point** or **critical number** of f if
(i) $f'(x_0) = 0$ or
(ii) $f'(x_0)$ does not exist.
If $f'(x_0) = 0$, then x_0 is also called a **stationary point**. Graphically, this means the graph has a horizontal tangent at $x = x_0$.



Horizontal tangent: $f'(x_0) = 0$



Corner: derivative does not exist

Example 3: Critical points

Find the critical points of

$$f(x) = x^3 - 3x + 1.$$

Answer.

Given

$$f(x) = x^3 - 3x + 1.$$

Differentiate:

$$f'(x) = 3x^2 - 3.$$

For critical points, solve

$$f'(x) = 0.$$

Thus

$$\begin{aligned} 3x^2 - 3 &= 0 \\ 3(x^2 - 1) &= 0 \\ x^2 - 1 &= 0 \\ (x-1)(x+1) &= 0. \end{aligned}$$

Therefore,

$$x = 1 \quad \text{or} \quad x = -1.$$

So the critical numbers are

$$\boxed{x = -1,\ 1}.$$

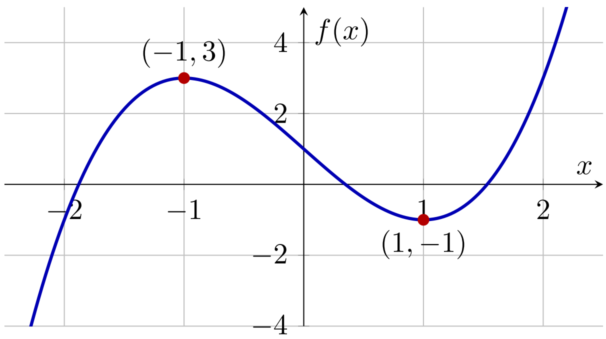
The corresponding points are

$$f(-1) = (-1)^3 - 3(-1) + 1 = -1 + 3 + 1 = 3,$$

$$f(1) = 1^3 - 3(1) + 1 = 1 - 3 + 1 = -1.$$

Hence the critical points on the graph are

$$\boxed{(-1, 3) \quad \text{and} \quad (1, -1)}.$$



The graph has horizontal tangents at the two critical points.

Example 04:

Find the critical points of $f(x) = 3x^{5/3} - 15x^{2/3}$.

Given

$$f(x) = 3x^{5/3} - 15x^{2/3}.$$

Differentiating with respect to x, we get

$$\begin{aligned} f'(x) &= 3 \cdot \frac{5}{3} x^{2/3} - 15 \cdot \frac{2}{3} x^{-1/3} \\ &= 5x^{2/3} - 10x^{-1/3} \\ &= \frac{5x - 10}{x^{1/3}} \\ &= \frac{5(x-2)}{x^{1/3}}. \end{aligned}$$

Critical points occur when $f'(x) = 0$ or $f'(x)$ does not exist.

$$f'(x) = 0 \Rightarrow \frac{5(x-2)}{x^{1/3}} = 0 \Rightarrow x = 2.$$

Also, $f'(x)$ does not exist at $x = 0$, but $f(0)$ exists. Therefore the critical points are

$$x = 0, \qquad x = 2.$$

Now we examine the sign of $f'(x)$.

Interval	Sign of $f'(x) = \dfrac{5(x-2)}{x^{1/3}}$	Behavior of f
$(-\infty, 0)$	$+$	Increasing
$(0, 2)$	$-$	Decreasing
$(2, \infty)$	$+$	Increasing

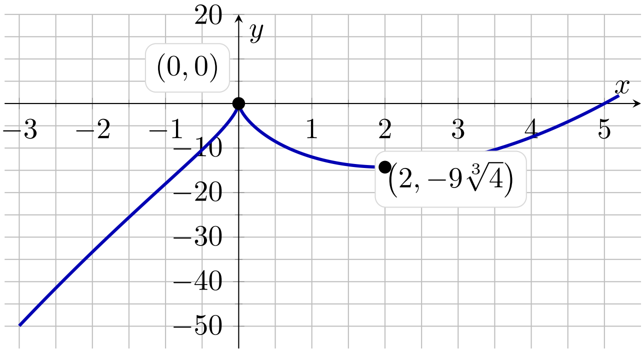


Figure: Graph of $f(x) = 3x^{5/3} - 15x^{2/3}$.

5 Practice Problems

Practice problems from the lecture

Solve the following exercises for more practice:

- Exercise 3.5: Problems 5–10 and 11–16.
- Exercise 10.1: Problems 53, 54 and 45–52.
- Exercise 4.1: Problems 15–32 (a,b).
- Exercise 4.2: Problems 7–14, 25–32 and 33–38.

6 Practice Problems

1. Find (a) the open intervals on which f is increasing, (b) the open intervals on which f is decreasing, (c) the open intervals on which f is concave up, (d) the open intervals on which f is concave down and (e) the x-coordinate of all inflection points.

   (i) $f(x) = x^2 - 5x + 6$ $\quad$ (ii) $f(x) = 5 + 12x - x^3$

   (iii) $f(x) = x^4 - 8x^2 + 16$ $\quad$ (iv) $f(x) = \dfrac{x^2}{x^2+2}$ $\quad$ (v) $f(x) = \sqrt[3]{x+2}$.

2. Locate the critical numbers and identify which critical numbers correspond to stationary points.

   (i) $f(x) = x^3 + 3x^2 - 9x + 1$ $\quad$ (ii) $f(x) = x^4 - 6x^2 - 3$

   (iii) $f(x) = \dfrac{x}{x^2+2}$ $\quad$ (iv) $f(x) = x^{2/3}$

   (v) $f(x) = x^{1/3}(x+4)$ $\quad$ (vi) $f(x) = \cos 3x$.